NeuMathe 题库导出

共 26 道题目 | 2025-12-19 17:02

1.

设 $u=\sqrt{x^2+y^2+z^2}$，则 $\frac{\partial^2 u}{\partial x^2}=$（ ）.

A $\frac{x^2+z^2}{x^2+y^2+z^2}$

B $\frac{y^2+z^2}{x^2+y^2+z^2}$

C $\frac{x^2+z^2}{\left(x^2+y^2+z^2\right)^{3/2}}$;

D $\frac{y^2+z^2}{\left(x^2+y^2+z^2\right)^{3/2}}$; 【答案】

【解析】

令 $r=u=\sqrt{x^2+y^2+z^2}$，$\frac{\partial u}{\partial x}=\frac{x}{r}$，$\frac{\partial^2 u}{\partial x^2}=\frac{r-x\frac{x}{r}}{r^2}=\frac{y^2+z^2}{r^3}=\frac{y^2+z^2}{\left(x^2+y^2+z^2\right)^{3/2}}$.

2.

设 $u=\frac{x}{y^2}+\ln xy$，则 $\frac{\partial^2 u}{\partial x\partial y}=$（ ）.

A $-\frac{2}{y^3}+\frac{1}{xy^2}$

B $\frac{2}{y^3}+\frac{1}{xy^2}$

C $-\frac{2}{y^3}-\frac{1}{xy^2}$

D $-\frac{2}{y^3}$ 【答案】

【解析】

$\frac{\partial u}{\partial x}=\frac{1}{y^2}+\frac{1}{x}$，$\frac{\partial^2 u}{\partial x\partial y}=-\frac{2}{y^3}$

3.

设 $u(x,y)=f(e^x)g(\sin y)$，其中 $f(x),g(x)$ 均有连续导数，则 $\frac{\partial^2 u}{\partial x\partial y}=()$

A $e^x\cos yf'(e^x)g'(\sin y)$ 【答案】

B $ue^x\cos yf'(e^x)g'(\sin y)$

C $ue^x\sin yf'(e^x)g'(\sin y)$

D $e^x\sin yf'(e^x)g'(\sin y)$

【解析】

$\frac{\partial u}{\partial x}=e^xf'(e^x)g(\sin y)$，则 $\frac{\partial^2 u}{\partial x\partial y}=e^x\cos yf'(e^x)g'(\sin y)$

4.

设 $u=e^{-x}\cos y$，则 $\left(\frac{\partial u}{\partial x}\right)^2+\frac{\partial^2 u}{\partial y^2}=$（ ）.

(A) $e^{-2x}\cos^2 y-e^{-x}\cos y$ 【答案】

(B) $e^{-2x}\cos y-e^{-x}\sin y$

(C) e^{-x}

(D) $e^{-x}(\sin y-\cos y)$

【解析】

$$\frac{\partial u}{\partial x}=-e^{-x}\cos y,\quad \frac{\partial u}{\partial y}=-e^{-x}\sin y,\quad \frac{\partial^2 u}{\partial y^2}=-e^{-x}\cos y,$$

则$\left(\frac{\partial u}{\partial x}\right)^2+\frac{\partial^2 u}{\partial y^2}=e^{-2x}\cos^2 y-e^{-x}\cos y.$

5.

设 $z=e^{-x}\sin\left(\frac{x}{y}\right)$，则 $\left.\frac{\partial^2 z}{\partial x\partial y}\right|_{\left(2,\frac{1}{\pi}\right)}=$().

(A) $\frac{3\pi}{4e}$.

(B) $\frac{\pi}{e}$

(C) $\frac{2\pi}{3e}$

(D) $\left(\frac{\pi}{e}\right)^2$ 【答案】

【解析】

$$\frac{\partial z}{\partial x}=-e^{-x}\sin\left(\frac{x}{y}\right)+\frac{e^{-x}}{y}\cos\left(\frac{x}{y}\right),$$

$$\frac{\partial^2 z}{\partial x\partial y}=-e^{-x}\left(-\frac{x}{y^2}\right)\cos\frac{x}{y}-\frac{e^{-x}}{y^2}\cos\frac{x}{y}+\frac{e^{-x}}{y}\left(-\frac{x}{y^2}\right)\left(-\sin\frac{x}{y}\right),$$

则$\left.\frac{\partial^2 z}{\partial x\partial y}\right|_{\left(2,\frac{1}{\pi}\right)}=\left(\frac{\pi}{e}\right)^2.$

6.

设 $z=\ln(e^x+e^y)$，则 $\frac{\partial^2 z}{\partial x\partial y}=$（ ）

A $\frac{-1}{(e^x+e^y)^2}$.

B $-\frac{e^y}{(e^x+e^y)^2}$;

C $-\frac{e^{x+y}}{(e^x+e^y)^2}$ 【答案】

D $-\frac{e^x}{(e^x+e^y)^2}$;

【解析】

$\frac{\partial z}{\partial x}=\frac{e^x}{e^x+e^y}$，$\frac{\partial^2 z}{\partial x\partial y}=-\frac{e^{x+y}}{(e^x+e^y)^2}$

7.

设 $z=e^x\sin y+e^{-y}\cos x$，则 $\frac{\partial^2 z}{\partial x\partial y}=()$.

A $e^y\sin y+e^{-x}\cos x$

B $e^x\sin y+e^{-y}\cos x$

C $e^y\cos x+e^{-x}\sin y$

D $e^x\cos y+e^{-y}\sin x$ 【答案】

【解析】

$\frac{\partial z}{\partial x}=e^x\sin y-e^{-y}\sin x$，则 $\frac{\partial^2 z}{\partial x\partial y}=e^x\cos y+e^{-y}\sin x$

8.

设 $f(x,y,z)=xy^3+yz^2+zx^2$，则 $f_{zzx}(2,0,1)=()$.

A 0 ； 【答案】

B 1;

C 3.

D 2 ；

【解析】

$f_z(x,y,z)=2yz+x^2$，$f_{zz}(x,y,z)=2y$，$f_{zzx}(x,y,z)=0$.

9.

设 $u(x,y)=f(\sin x)g(e^y)$，其中 $f(x),g(x)$ 均有连续的导数，则 $\frac{\partial^2 u}{\partial x\partial y}=$()

(A) $f'(\sin x)g'(e^y)\cos xe^y$; 【答案】

(B) $f'(\sin x)g'(e^y)\cos x$

(C) $uf'(\sin x)g'(e^y)$;

(D) $f'(\sin x)g'(e^y)$;

【解析】

$\frac{\partial u}{\partial x}=f'(\sin x)g(e^y)\cos x,\quad \frac{\partial^2 u}{\partial x\partial y}=f'(\sin x)g'(e^y)\cos xe^y$

10.

设 $u=x^2-2bxy-cy^2,\left.\frac{\partial u}{\partial y}\right|_{(2,1)}=0$，则 $\frac{\partial^2 u}{\partial y^2}=$().

(A) $2b$

(B) $-4b$

(C) $-2b$

(D) $4b$ 【答案】

【解析】

$\frac{\partial u}{\partial y}=-2bx-2cy,\quad \left.\frac{\partial u}{\partial y}\right|_{(2,1)}=-4b-2c=0, c=-2b$，则 $\frac{\partial^2 u}{\partial y^2}=-2c=4b$

11.

设 $u=x^2\sin y+y^2\cos x$ ，则 $\frac{\partial^2 u}{\partial x\partial y}=$().

(A) $xy\sin y-xy\cos x$

(B) $2x\cos y-2y\sin x$ 【答案】

(C) $x^2\cos y+y^2\sin x$

(D) $x^2\cos y+y\sin x$

【解析】

$\frac{\partial u}{\partial x}=2x\sin y-y^2\sin x,\quad \frac{\partial^2 u}{\partial x\partial y}=2x\cos y-2y\sin x$

12.

设 $u=\arctan\frac{x}{y}$ ，则 $\frac{\partial^2 u}{\partial x\partial y}=$

A $\frac{1}{x^2+y^2}$

B $\frac{x^2-y^2}{x^2+y^2}$

C $\frac{x^2+3y^2}{\left(x^2+y^2\right)^2}$;

D $\frac{x^2-y^2}{\left(x^2+y^2\right)^2}$ 【答案】

【解析】

$\frac{\partial u}{\partial x}=\frac{1}{1+\left(\frac{x}{y}\right)^2}\cdot\frac{1}{y}=\frac{y}{x^2+y^2}$，$\frac{\partial^2 u}{\partial x\partial y}=\frac{x^2+y^2-y\cdot 2y}{\left(x^2+y^2\right)^2}=\frac{x^2-y^2}{\left(x^2+y^2\right)^2}$

13.

设 $u=xy+\frac{y}{x}$，则 $\frac{\partial^2 u}{\partial x^2}=$（ ）.

A $\frac{2y}{x^3}$ 【答案】

B $\frac{y}{3x^2}$

C $\frac{y^3}{2x}$.

D $\frac{y^2}{2x}$

【解析】

$\frac{\partial u}{\partial x}=y-\frac{y}{x^2}$，则 $\frac{\partial^2 u}{\partial x^2}=\frac{2y}{x^3}$

14.

设 $f(x,y,z)=xy^2+yz^2+zx^2$，则下列偏导数的计算中错误的是（ ）.

A $\frac{\partial^2 f}{\partial x\partial z}(1,0,2)=2;$

B $\frac{\partial^2 f}{\partial y\partial z}(0,-1,0)=0;$

C $\frac{\partial^3 f}{\partial z^2\partial x}(1,1,1)=0$

D $\frac{\partial^2 f}{\partial x^2}(0,0,1)=0;$ 【答案】

【解析】

$\frac{\partial f}{\partial x}=y^2+2xz,\frac{\partial^2 f}{\partial x^2}=2z,\frac{\partial^2 f}{\partial x\partial z}=2x,\frac{\partial f}{\partial y}=2xy+z^2,\frac{\partial^2 f}{\partial y\partial z}=2z,\frac{\partial f}{\partial z}=2yz+x^2,\frac{\partial^2 f}{\partial z^2}=2y,$
$\frac{\partial^3 f}{\partial z^2\partial x}=0,$ 所以$\frac{\partial^2 f}{\partial x^2}(0,0,1)=2,\frac{\partial^2 f}{\partial x\partial z}(1,0,2)=2,\frac{\partial^2 f}{\partial y\partial z}(0,-1,0)=0,$
$\frac{\partial^3 f}{\partial z^2\partial x}(1,1,1)=0.$

15.

设 $z=e^{-\left(\frac{1}{x}+\frac{1}{y}\right)}$ ，则下列结论错误的是（ ）。

A $\frac{\partial^2 z}{\partial x \partial y}=\frac{e^{-\left(\frac{1}{x}+\frac{1}{y}\right)}}{x^2 y^2}$

B $x^2 \frac{\partial z}{\partial x}+y^2 \frac{\partial z}{\partial y}=z$ 【答案】

C $\frac{\partial^2 z}{\partial x^2}=\frac{(1-2x)e^{-\left(\frac{1}{x}+\frac{1}{y}\right)}}{x^4}$

D $\frac{\partial^2 z}{\partial y^2}=\frac{(1-2y)e^{-\left(\frac{1}{x}+\frac{1}{y}\right)}}{y^4}$

【解析】

$\frac{\partial z}{\partial x}=\frac{e^{-\left(\frac{1}{x}+\frac{1}{y}\right)}}{x^2}, \frac{\partial z}{\partial y}=\frac{e^{-\left(\frac{1}{x}+\frac{1}{y}\right)}}{y^2}$, 则$x^2 \frac{\partial z}{\partial x}+y^2 \frac{\partial z}{\partial y}=2z, \frac{\partial^2 z}{\partial x^2}=\frac{(1-2x)e^{-\left(\frac{1}{x}+\frac{1}{y}\right)}}{x^4}$,
$\frac{\partial^2 z}{\partial y^2}=\frac{(1-2y)e^{-\left(\frac{1}{x}+\frac{1}{y}\right)}}{y^4}, \frac{\partial^2 z}{\partial x \partial y}=\frac{e^{-\left(\frac{1}{x}+\frac{1}{y}\right)}}{x^2 y^2}$.

16.

设 $z=\ln \sqrt{x^2+y^2}$，则 $\frac{\partial^2 z}{\partial x^2}+\frac{\partial^2 z}{\partial y^2}=()$.

A 0 ； 【答案】

B $\frac{1}{x^2+y^2}$;

C $\frac{y^2}{\left(x^2+y^2\right)^2}$

D $\frac{x^2}{\left(x^2+y^2\right)^2}$;

【解析】

$\frac{\partial z}{\partial x}=\frac{1}{\sqrt{x^2+y^2}} \cdot \frac{1}{2} \frac{2x}{\sqrt{x^2+y^2}}=\frac{x}{x^2+y^2}, \frac{\partial z}{\partial y}=\frac{1}{\sqrt{x^2+y^2}} \cdot \frac{1}{2} \frac{2y}{\sqrt{x^2+y^2}}=\frac{y}{x^2+y^2}$,
$\frac{\partial^2 z}{\partial x^2}=\frac{x^2+y^2-x \cdot 2x}{\left(x^2+y^2\right)^2}=\frac{y^2-x^2}{\left(x^2+y^2\right)^2}, \quad \frac{\partial^2 z}{\partial y^2}=\frac{x^2+y^2-y \cdot 2y}{\left(x^2+y^2\right)^2}=\frac{x^2-y^2}{\left(x^2+y^2\right)^2}$,
则$\frac{\partial^2 z}{\partial x^2}+\frac{\partial^2 z}{\partial y^2}=0$.

17.

设 $u=\frac{x}{y^2}+\frac{y}{x}$，则 $\frac{\partial^2 u}{\partial x^2}=()$.

A $\frac{2y^2}{3x^3}$

B $\frac{y^3}{2x^2}$

C $\frac{2y}{x^3}$ 【答案】

D $\frac{2y}{5x^2}$

【解析】

$\frac{\partial u}{\partial x}=\frac{1}{y^2}-\frac{y}{x^2}$，则 $\frac{\partial^2 u}{\partial x^2}=\frac{2y}{x^3}$

18.

设 $f(x,y,z)=xy^2+yz^2+zx^2$，则 $f_{xz}(1,0,2)=()$.

A 2 【答案】

B 1

C 0

D -1

【解析】

$f_x(x,y,z)=y^2+2zx, f_{xz}(x,y,z)=2x$，则 $f_{xz}(1,0,2)=2$

19.

设 $z=\ln\left(xy^2\right)$ ， 下列偏导数的计算错误的是（ ）。

A $\frac{\partial^2 z}{\partial x^2}=-\frac{1}{x^2}$

B $\frac{\partial^2 z}{\partial x \partial y}=0$

C $\frac{\partial^2 z}{\partial x \partial y}=\frac{1}{x}$ 【答案】

D $\frac{\partial^2 z}{\partial y^2}=-\frac{2}{y^2}$

【解析】

$\frac{\partial z}{\partial x}=\frac{1}{xy^2}\cdot y^2=\frac{1}{x}$，$\frac{\partial z}{\partial y}=\frac{1}{xy^2}\cdot 2xy=\frac{2}{y}$，$\frac{\partial^2 z}{\partial x^2}=-\frac{1}{x^2}$，$\frac{\partial^2 z}{\partial y^2}=-\frac{2}{y^2}$，$\frac{\partial^2 z}{\partial x \partial y}=0$.

20.

设 $u=x\ln xy$，则 $\frac{\partial^2 u}{\partial x\partial y}=()$.

A $\frac{1}{x}$

B $\frac{x}{y}$

C $\frac{y}{x}$

D $\frac{1}{y}$ 【答案】

【解析】

$\frac{\partial u}{\partial x}=\ln xy+1$，则 $\frac{\partial^2 u}{\partial x\partial y}=\frac{1}{y}$

21.

设 $u=xyz+\frac{xy}{z}+\frac{zx^2}{y}+\frac{yz}{x}$，则 $\frac{\partial^2 u}{\partial x^2}=(\quad)$.

A $\frac{2z}{y}+yz$

B $\frac{2z}{y}+2\frac{yz}{x^3}$ 【答案】

C $\frac{z}{y}+2\frac{yz}{x^3}$

D $\frac{2zx}{y}-2\frac{yz}{x^3}$

【解析】

解答 $\frac{\partial u}{\partial x}=yz+\frac{y}{z}+\frac{2zx}{y}-\frac{yz}{x^2}$, $\frac{\partial^2 u}{\partial x^2}=\frac{2z}{y}+2\frac{yz}{x^3}$.

22.

设 $u=x^4+y^4-4x^2y^2$，则 $\frac{\partial^2 u}{\partial y^2}=$ （ \quad ）.

A $\frac{\partial^2 u}{\partial y^2}=12x^2-16xy$

B $12x^2-8y^2$;

C $\frac{\partial^2 u}{\partial y^2}=12y^2-16xy$

D $12y^2-8x^2$; 【答案】

【解析】

$\frac{\partial u}{\partial y}=4y^3-8x^2y,\quad \frac{\partial^2 u}{\partial y^2}=12y^2-8x^2$

23.

下面结论中错误的是（ ）.

Ⓐ $z=f(x,y)$ 在 $P_0(x_0,y_0)$ 连续，且 $f_x(x_0,y_0),f_y(x_0,y_0)$ 存在，则函数在 $P_0(x_0,y_0)$ 必可微； 【答案】

Ⓑ $z=f(x,y)$ 两个混合偏导数 $\frac{\partial^2 z}{\partial x\partial y},\frac{\partial^2 z}{\partial y\partial x}$ 连续，则必有 $\frac{\partial^2 z}{\partial x\partial y}=\frac{\partial^2 z}{\partial y\partial x}$；

Ⓒ $z=f(x,y)$ 在 $P_0(x_0,y_0)$ 连续，则必有 $\lim\limits_{\substack{x\to x_0\\y\to y_0}}f(x,y)$ 存在；

Ⓓ 若 $z=f(x,y)$ 在 $P(x,y)$ 处可微，且知道 $dz=3x^2y^2dx+2x^3ydy$，则必有 $f(x,y)=x^3y^2+C$.

【解析】

函数 $f(x,y)=\begin{cases}\frac{xy}{\sqrt{x^2+y^2}}, & x^2+y^2\neq 0,\\ 0, & x^2+y^2=0,\end{cases}$ 在原点处连续且 $f_x(0,0),f_y(0,0)$ 存在，但是不可微。

24.

函数 $z=f(x,y)$ 在点 (x,y) 处的二阶偏导数 $f_{xy}(x,y)$ 及 $f_{yx}(x,y)$ 都存在，则 $f_{xy}(x,y)$ 及 $f_{yx}(x,y)$ 在点 (x,y) 处连续是 $f_{xy}=f_{yx}$ 的（ ）.

Ⓐ 充分而非必要条件； 【答案】

Ⓑ 必要而非充分条件；

Ⓒ 充分必要条件；

Ⓓ 既非充分又非必要条件.

【解析】

解答 考虑 $f(x,y)=\begin{cases}\frac{x^2y^2}{x^2+y^2}, & (x,y)\neq(0,0),\\ 0, & (x,y)=(0,0).\end{cases}$

可求得

$$f_x(x,y)=\begin{cases}\frac{2xy^4}{(x^2+y^2)^2}, & (x,y)\neq(0,0),\\ 0, & (x,y)=(0,0),\end{cases}\quad f_y(x,y)=\begin{cases}\frac{2x^4y}{(x^2+y^2)^2}, & (x,y)\neq(0,0),\\ 0, & (x,y)=(0,0).\end{cases}$$

$$f_{xy}=\begin{cases}\frac{8x^3y^3}{(x^2+y^2)^3}, & (x,y)\neq(0,0),\\ 0, & (x,y)=(0,0),\end{cases}\quad f_{yx}=\begin{cases}\frac{8x^3y^3}{(x^2+y^2)^3}, & (x,y)\neq(0,0),\\ 0, & (x,y)=(0,0).\end{cases}$$

则 $f_{xy}=f_{yx}$. 但是，当 $(x,y)\to(0,0)$，沿直线 $y=kx$ 有

$$\lim_{(x,y)\to(0,0)}f_{xy}(x,y)=\lim_{\substack{x\to 0\\y=kx}}f_{xy}(x,y)=\lim_{\substack{x\to 0\\y=kx}}\frac{8k^3x^6}{\left(1+k^2\right)^3x^6}=\frac{8k^3}{\left(1+k^2\right)^3}$$

可见 $\lim\limits_{(x,y)\to(0,0)}f_{xy}(x,y)$ 不存在，因此 $f_{xy}(x,y)$ 在 $(0,0)$ 点不连续.

25.

设 $z=\left(x+y^2\right)\sin(xy)$，则 $\frac{\partial^2 z}{\partial x^2}=()$

A $2y\cos(xy)-\left(xy+y^3\right)\sin(xy)$.

B $y\cos(xy)-\left(xy^2+y^4\right)\sin(xy)$;

C $2y\cos(xy)-\left(x+y^2\right)\sin(xy)$;

D $2y\cos(xy)-\left(xy^2+y^4\right)\sin(xy)$; 【答案】

【解析】

$\frac{\partial z}{\partial x}=\sin(xy)+\left(x+y^2\right)y\cos(xy)$

$\frac{\partial^2 z}{\partial x^2}=y\cos(xy)+y\cos(xy)-\left(x+y^2\right)y^2\sin(xy)=2y\cos(xy)-\left(xy^2+y^4\right)\sin(xy)$

26.

设 $f(x,y)=x^2+\left(y^2-1\right)\tan\sqrt{\frac{x}{y}}$，则 $f_x(x,1)=(\quad)$.

A $3x$

B 0

C x

D $2x$ 【答案】

【解析】

$f(x,1)=x^2$，则 $f_x(x,1)=2x$.